BookBridge TSA3 — service and domain boundaries

Application
BookBridgeApp: console demonstration

Automated tests
Service, polymorphism, legacy regression

Exception package
DuplicateEntity / ItemUnavailable
LoanLimitExceeded (duration)

Service: LibraryService
Validation, circulation, queue fairness
Search, sorting, aggregates; no UI or DB

Helpers and injected inputs
Generic Lookup; Validation
Clock, Supplier<String> IDs

Service-owned collections
LinkedHashMap: books, members, active loans
ArrayDeque: pending reservations
HashSet: checked-out ISBNs, used IDs
ArrayList: loan history

Integrity boundaries
Validate and construct before mutation.
Immutable collection and domain snapshots.
FIFO per ISBN; skip unavailable books.
In memory; single-threaded; no persistence.

Model package: immutable values and polymorphic policies

LibraryBook
ISBN, title, BookStatus

LoanRecord
ID, ISBN, Member, dates

Reservation
ID, member ID, ISBN, duration

Notifiable interface
notificationMessage(): String

Abstract Member
Final ID/name; polymorphic maximum days and daily fine
LoanRecord references Member; subclasses extend Member

StudentMember
14 days / PHP 5

FacultyMember
30 days / PHP 2

GuestMember
7 days / PHP 10

Enums: BookStatus, LoanStatus, MembershipType, RequestOutcome
CirculationSummary: immutable counts, membership groups, estimated fines

Arrows show dependencies / interface implementation / inheritance. Swing and JDBC are deferred to TSA4.